

HEALTHCONNECT CLINIC EXPERIENCE LAB

Project Title: Improving Patient Appointment Attendance and Healthcare Support Using Data and AI

Week 7: Testing, Refinement and End-to-End Validation

WEEK 6 TO WEEK 7 TRANSITION

In Week 6, the analysis moved beyond the initial descriptive analysis by investigating relationships between previous no-shows, booking lead time, reminders and other appointment characteristics. The main output was an improved Power BI analytical dashboard containing segmented analysis, a no-show risk heatmap and an evidence table.

The most important Week 6 result was the identification of stronger no-show patterns among patients with previous no-shows and appointments booked far in advance. For example, appointments with **31+ booking lead days had a 60.49% no-show rate**, while appointments involving patients with **2 or more previous no-shows had a 61.02% no-show rate**. The combined segment of patients with 2+ previous no-shows and 31+ booking lead days recorded a **72.96% no-show rate**.

Week 6 also identified limitations that required further testing, including the interpretation of the high-risk segment, reminder effectiveness, dashboard calculation accuracy and the reliability of some segments such as Sunday appointments.

Therefore, Week 7 focused on testing the analytical calculations, validating the dashboard against the underlying dataset, testing filters, checking important findings across segments, refining the dashboard and validating relevant outputs with other HealthConnect tracks.

WEEK 7 TESTING READINESS

Component Being Tested

The main component tested in Week 7 was the **Week 6 Power BI analytical dashboard and its underlying KPI and segmentation calculations**.

The component was designed to:

- Calculate accurate appointment KPIs.
- Identify patterns associated with missed appointments.
- Allow users to investigate important patient and appointment segments.
- Communicate high-risk no-show patterns clearly.
- support evidence-based recommendations for improving attendance and patient support.

The main connected track was **Data Analytics**, with cross-track validation involving **Data Science** because the validated analytical findings and variables can support future predictive modelling.

Testing Objective

The Week 7 testing aimed to determine whether:

1. The main KPIs were calculated correctly.
2. Power BI dashboard values matched the underlying dataset.
3. Dashboard filters produced consistent results.
4. Important Week 6 findings remained valid when independently checked.
5. The high-risk segment was correctly defined.
6. The dashboard communicated the analytical findings clearly.
7. Important variables remained useful for further HealthConnect development.
8. The refined outputs could be reliably shared with other HC-POD tracks.

TESTING APPROACH

The testing was performed using several methods.

First, the main KPI calculations were checked against the underlying 5,000 appointment records. The total number of appointments, attended appointments, no-shows and cancellations were independently verified.

Second, important segmented calculations were checked using the dataset and compared with Power BI results. This included previous no-show groups, booking lead-time groups and combined risk segments.

Third, dashboard filters and slicers were tested to ensure that changing variables such as appointment type, gender, age group, appointment day and reminder status did not produce unexpected results.

Fourth, Week 6 findings were re-tested to determine whether they remained valid rather than simply accepting the original results.

Finally, the dashboard and validated analytical findings were reviewed for clarity and usefulness, leading to refinements where necessary.

KPI VALIDATION

The main KPIs were tested against the underlying HealthConnect appointment dataset.

The dataset contains 5,000 appointment records.

The validation confirmed: Total Appointments = 5,000, Attended Appointments = 2,314, No-Show Appointments = 2,423, Cancelled Appointments = 263, Attendance Rate = 46.28%, No-Show Rate = 48.46%, Cancellation Rate = 5.26%. Reminder Coverage = 72.68%.

The KPI calculations passed validation because the Power BI results agreed with the underlying appointment records.

The outcome totals also reconcile correctly: $2,314 + 2,423 + 263 = 5,000$ appointments. This confirmed that the three appointment outcomes were being counted correctly.

ANALYTICAL VALIDATION

The Week 6 findings were re-tested instead of being assumed to be correct.

Previous No-Shows

The analysis continued to show an increasing no-show pattern as previous no-shows increased.

The validated no-show rates were: 0 previous no-shows: 43.51%, 1 previous no-show: 53.49%, 2 previous no-shows: 59.36%, 3+ previous no-shows: 68.82%.

This confirms that previous no-show history is an important indicator for appointment attendance analysis.

However, this relationship should still be interpreted as an association rather than proof of causation.

Booking Lead Time

The booking lead-time analysis was also re-tested.

The validated no-show rates were: 0–1 days: **26.51%**, 2–7 days: **28.27%**, 8–14 days: **33.55%**, 15–30 days: **43.21%**, 31+ days: **60.49%**.

The result remained consistent with the Week 6 finding that appointments booked far in advance had substantially higher no-show rates in this dataset.

Combined Risk Segment

The combined segment was also checked.

Appointments with: **2+ previous no-shows**, and **31+ booking lead days** contained **233 appointments**, of which **170 were no-shows**, producing a no-show rate of **72.96%**.

This was important because it clarified the difference between the two Week 6 percentages.

The **61.02%** figure represents the broader group with **2+ previous no-shows**, while **72.96%** represents the narrower combined segment of **2+ previous no-shows and 31+ booking lead days**.

Therefore, the Week 7 report uses **72.96%** when referring specifically to the combined high-risk segment.

KPI AND DASHBOARD TESTING

The Power BI dashboard was tested using the major slicers and filters.

The date filter was tested to confirm that the KPIs changed according to the selected appointment period.

Appointment type was tested to confirm that the dashboard correctly recalculated the outcomes for General Consult, Specialist Consult, Follow-up and Diagnostic Test appointments.

Gender and age-group filters were also tested to confirm that the displayed appointment counts and outcome rates responded to the selected population.

Appointment day and reminder status were tested because these variables were important to the Week 6 investigation.

The testing showed that the dashboard responded appropriately to the selected filters and maintained consistent KPI calculations.

No calculation error requiring a complete rebuild of the dashboard was identified.

DASHBOARD REFINEMENT

Although the Week 6 dashboard was functional, Week 7 focused on making the analytical message clearer.

The main refinements were:

1. Clearer High-Risk Definition

The high-risk segment was explicitly defined as:

Previous No-Shows ≥ 2 AND Booking Lead Days ≥ 31 .

This prevents confusion between the broader 2+ previous no-show group and the combined high-risk group.

2. Improved Evidence Display

The evidence table was retained because it provides the actual appointment count, no-show count and no-show rate behind the important findings.

This makes the dashboard easier to validate and reduces the possibility of interpreting percentages without considering segment size.

3. Improved Heatmap Interpretation

The Week 6 matrix comparing previous no-shows with booking lead time was retained because it provides a useful view of how the two factors interact.

The heatmap helps identify where no-show rates become particularly high rather than looking at either variable separately.

4. Reduced Unnecessary Visual Emphasis

Visuals that repeated information already communicated by the main KPIs were given less emphasis so that the dashboard could focus attention on the most actionable patterns.

5. Clearer Business Interpretation

The dashboard was refined to distinguish between overall no-show rate, individual risk indicators; and the combined high-risk segment.

This makes the dashboard more useful for decision-making.

REMINDER EFFECTIVENESS VALIDATION

Reminder status was re-tested because it was one of the variables investigated in Week 5 and Week 6.

The validated results were: No reminder: **51.39% no-show rate** and Reminder sent: **47.36% no-show rate**. This shows a lower observed no-show rate among appointments where a reminder was sent. However, the result does **not establish that reminders caused the reduction**. Other differences between patients or appointments may contribute to the observed association.

Therefore, the Week 7 recommendation is not to claim that reminders definitely prevent no-shows. Instead, the result supports continued investigation of reminder coverage, timing and channel effectiveness.

VALIDATION OF OTHER IMPORTANT FINDINGS

The Week 6 analysis identified distance and appointment type as additional variables worth monitoring. Appointments more than 20 km from the clinic had a **57.76% no-show rate**, compared with **45.28%** for appointments 2–5 km from the clinic. This supports distance as a relevant access-related indicator in the dataset.

Appointment type also showed differences, with Follow-up appointments recording a **51.23% no-show rate**, Diagnostic Tests **49.75%**, Specialist Consultations **47.44%**, and General Consultations **46.64%**.

These differences were retained as descriptive findings rather than treated as causal explanations.

The waiting-time variable was not elevated as a major risk factor because 60 records contained missing waiting-time values and the Week 6 analysis did not identify a strong reliable pattern.

CROSS-TRACK TESTING AND REFINEMENT

The main cross-track collaboration for Week 7 was between Data Analytics and Data Science.

Project Dependency

The Data Science track requires reliable analytical variables and validated patterns for future predictive modelling.

Component Tested

The variables and analytical findings identified by Data Analytics as potentially useful for modelling were reviewed for consistency and analytical relevance.

These included: previous no-shows, booking lead time, reminder status, reminder channel, appointment type, distance to clinic.

Information Provided

Data Analytics provided the validated findings showing that previous no-shows and long booking lead times were associated with substantially higher no-show rates.

The combined 2+ previous no-show and 31+ lead-time segment was also provided as a validated analytical segment.

Testing Activity

The findings were reviewed as potential predictive features while checking that they represented information available before or around the appointment rather than information created after the outcome.

Finding

The testing highlighted the importance of distinguishing between **historical information available before an appointment** and variables that could introduce data leakage if they were only known after the appointment outcome.

Action Taken

The validated pre-appointment variables were retained as potential inputs for predictive analysis, while variables that could potentially contain post-outcome information were treated cautiously.

Retest Result

The refined analytical outputs remained consistent with the original dataset calculations.

Resulting Improvement

The collaboration strengthened the connection between descriptive analytics and future predictive modelling by providing Data Science with validated, clearly defined analytical variables.

This represents meaningful cross-track collaboration because it resulted in a **test** → **finding** → **action** → **retest** process rather than communication alone.

TESTING, REFINEMENT AND VALIDATION RECORD

The Week 7 testing produced the following main outcomes in narrative form.

Component Tested	Expected Result	Actual Result	Status	Action
Total Appointments	5,000	5,000	PASS	None
No-Show Rate	48.46%	48.46%	PASS	None
Attendance Rate	46.28%	46.28%	PASS	None
Cancellation Rate	5.26%	5.26%	PASS	None
Lead 31+ No-Show Rate	60.49%	60.49%	PASS	None
Previous No-Shows 2+ Rate	61.02%	61.02%	PASS	None
Combined High-Risk Segment	72.96%	72.96%	PASS	Definition clarified

Dashboard Filters: The expected result was that KPIs and visuals would respond correctly to the selected filters. The actual result was consistent with the selected filters. The test passed.

Reminder Analysis: The expected result was to reproduce the Week 6 reminder comparison. The actual result remained 47.36% no-show for reminded appointments versus 51.39% for appointments without reminders. The test passed, with the limitation that the result represents association rather than causation.

CHART AND VISUALISATION TESTING

Appointment Outcome Chart

Tested: Attendance, No-Show and Cancelled counts. **Expected:** Total should equal 5,000. **Actual:** $2,314 + 2,423 + 263 = 5,000$. **Status:** PASS. **Action:** No change required.

2. Previous No-Show Group Chart Tested: No-show rates by previous no-shows. **Expected:** Rate should increase with previous no-shows. **Actual:** 43.51% to 68.82%. **Status:** PASS. **Action:** Retained.

Booking Lead-Time Chart

Tested: No-show rates by lead time. **Expected:** Rates should match validated results. **Actual:** 26.51% to 60.49%. **Status:** PASS. **Action:** Retained and labelled clearly.

Reminder Status Chart

Tested: No-show rates with and without reminders. **Expected:** Results should match the dataset. **Actual:** 47.36% with reminders vs 51.39% without. **Status:** PASS. **Action:** Clarified as an association, not causation.

Reminder Channel Chart

Tested: SMS, Email and WhatsApp results. **Expected:** Results should match the data. **Actual:** SMS 45.8%, Email 48.4%, WhatsApp 49.8%. **Status:** PASS. **Action:** Retained.

Appointment Type Chart

Tested: No-show rates by appointment type. **Expected:** Each type should show the correct rate. **Actual:** Rates ranged from 46.64% to 51.23%. **Status:** PASS. **Action:** Retained.

Distance Group Chart

Tested: No-show rates by distance. **Expected:** Rates should match the data. **Actual:** >20 km = 57.76%; 2–5 km = 45.28%. **Status:** PASS. **Action:** Retained.

Previous No-Show × Lead-Time Heatmap

Tested: Interaction between previous no-shows and lead time. **Expected:** Higher-risk combinations should be visible. **Actual:** Higher rates occurred with previous no-shows and longer lead times. **Status:** PASS. **Action:** Retained as the main analytical visual.

Previous No-Show × Reminder Chart

Tested: Reminder results by previous no-show group. **Expected:** Results should remain consistent. **Actual:** Results matched the analysis. **Status:** PASS. **Action:** Retained.

Booking Lead × Reminder Chart

Tested: Reminder results across lead-time groups. **Expected:** Filters should work correctly. **Actual:** Results updated correctly. **Status:** PASS. **Action:** Retained.

Distance × Reminder Chart

Tested: Reminder results across distance groups. **Expected:** Filters should update results correctly. **Actual:** Results updated correctly. **Status:** PASS. **Action:** Retained.

Risk Factor Count Chart

Tested: Number of risk indicators per appointment. **Expected:** Counts should be correct. **Actual:** Segmentation was reproduced correctly. **Status:** PASS. **Action:** Retained.

Long Lead + Distance Chart

Tested: Distance patterns for 31+ day bookings. **Expected:** Results should update correctly when filtered. **Actual:** Results updated correctly. **Status:** PASS. **Action:** Retained.

ISSUES IDENTIFIED AND RESOLVED

The most important analytical issue identified during Week 7 was the inconsistent use of the term **high-risk segment**.

The Week 6 analysis contained two different percentages:

61.02% for appointments involving patients with 2+ previous no-shows, and

72.96% for appointments involving patients with 2+ previous no-shows AND 31+ booking lead days.

The issue was resolved by defining the combined high-risk segment explicitly.

Therefore:

High-Risk Segment = Previous No-Shows $\geq$ 2 AND Booking Lead Days $\geq$ 31

and its validated no-show rate is:

72.96% based on 233 appointments and 170 no-shows.

The 61.02% figure remains valid but refers to the broader **2+ previous no-show** group.

REMAINING LIMITATIONS, RISKS AND DEPENDENCIES

Several limitations remain after testing.

The dataset is fictional and anonymised, so the findings should not automatically be assumed to represent real-world patient behaviour.

The analysis identifies associations and does not establish causation.

Reminder effectiveness cannot be interpreted as a causal effect because reminder assignment may be related to other patient or appointment characteristics.

The 60 missing waiting-time values limit the strength of conclusions about waiting time.

There are also 90 missing distance values, although the distance analysis still provides useful descriptive evidence.

The Sunday appointment issue remains a data-quality and operational dependency. The dataset contains Sunday appointments even though the HealthConnect Knowledge Base states that the clinic is closed on Sundays. Therefore, Sunday-related patterns should not be used as a major operational conclusion until the inconsistency is clarified.

The predictive use of analytical variables also requires continued attention to data leakage and the availability of each variable at prediction time.

UPDATED BUSINESS INSIGHTS

The Week 7 testing confirmed that the most important analytical patterns from Week 6 remained valid.

The first important finding is that previous no-show history is strongly associated with future missed appointments in the dataset. The no-show rate rises from **43.51%** for patients with no previous no-shows to **68.82%** among those with 3+ previous no-shows.

The second finding is that long booking lead time is associated with increased no-show rates. The rate rises from **26.51%** for appointments booked 0–1 days ahead to **60.49%** for appointments booked 31+ days ahead.

The third finding is that the combination of multiple previous no-shows and long booking lead time identifies a smaller but higher-risk segment. This group contains **233 appointments and has a 72.96% no-show rate**.

The fourth finding is that reminders are associated with a lower observed no-show rate, although the available data does not prove that reminders directly caused the difference.

The fifth finding is that distance from the clinic may represent an access-related factor, with appointments more than 20 km away showing a **57.76% no-show rate**.

UPDATED RECOMMENDATIONS

Based on the Week 7 validation, HealthConnect should consider **a targeted, non-punitive support approach** for patients with repeated previous no-shows.

Appointments booked 31+ days in advance should also receive appropriate follow-up and reminder support because this group recorded a 60.49% no-show rate.

Patients who fall into the combined high-risk segment should be considered for additional appointment support, such as timely reminders, confirmation opportunities and easier rescheduling information.

Reminder coverage should continue to be monitored, while further testing should be conducted to determine whether differences between reminder channels are meaningful.

Distance-related barriers should also be investigated because patients travelling longer distances showed higher observed no-show rates.

The Sunday data inconsistency should be resolved before operational decisions are made using Sunday appointment results.

These recommendations should be treated as decision-support actions based on observed patterns, not as proof that a particular intervention will cause attendance to increase.

END-TO-END HEALTHCONNECT VALIDATION

The Week 7 testing demonstrates that the Data Analytics component contributes to the wider HealthConnect solution.

The analytical component was tested by validating its KPIs, checking its segmented findings, testing dashboard interactions and refining the interpretation of important risk groups.

The expected outcome was that the dashboard would provide accurate and understandable evidence about appointment attendance and missed appointments.

The actual result was that the major KPIs and important analytical findings remained consistent after testing.

The main weakness identified was the unclear definition of the high-risk segment, which was corrected.

After refinement, the high-risk segment was clearly defined and retested, producing a validated 72.96% no-show rate for the 233 appointments meeting both criteria.

The Data Analytics outputs can therefore provide validated evidence to the wider HealthConnect solution, particularly for patient-support decisions and future predictive modelling.

WEEK 8 RECOMMENDATIONS

Before Week 8 final integration and presentation, the following activities should be completed:

1. Confirm the final Power BI dashboard and KPI calculations.
2. Maintain the corrected high-risk segment definition.
3. Resolve or clearly document the Sunday appointment inconsistency.
4. Confirm the final cross-track analytical variables with Data Science.
5. Ensure recommendations are presented as evidence-based associations rather than causal claims.
6. Finalise dashboard screenshots and testing evidence.
7. Ensure the final HealthConnect presentation uses the validated Week 7 figures.

WEEK 7 PROJECT SUMMARY

Week 7 moved the HealthConnect Data Analytics work from advanced analysis into systematic testing, refinement and end-to-end validation. The main component tested was the Week 6 Power BI analytical dashboard, including its KPIs, segmented analysis, filters and high-risk identification.

The KPI validation confirmed the main dataset totals and rates, including 5,000 appointments, a 48.46% no-show rate, 46.28% attendance rate and 5.26% cancellation rate.

The important Week 6 findings were also re-tested. Previous no-shows and long booking lead time remained the strongest observed indicators of missed appointments. The 31+ booking lead group recorded a 60.49% no-show rate, while patients with 2+ previous no-shows recorded a 61.02% rate.

A major refinement was made to the definition of the combined high-risk segment. The segment was clearly defined as patients with 2+ previous no-shows and 31+ booking lead days. The validated result was 233 appointments, 170 no-shows and a 72.96% no-show rate.

Dashboard filters and calculations were also tested and remained consistent with the underlying data. The dashboard was refined to improve clarity, reduce ambiguity and make the evidence behind important findings easier to understand.

Cross-track testing was conducted with Data Science by validating analytical variables and findings that can support future predictive modelling. The collaboration also highlighted the importance of avoiding data leakage when selecting predictive variables.

The main remaining limitations are the fictional nature of the dataset, missing distance and waiting-time values, the inability to establish causation from observational data, and the unresolved Sunday appointment inconsistency.

Overall, Week 7 confirmed that the Data Analytics component is functioning as intended after refinement and is ready to contribute to the final HealthConnect integration.